OFFICIAL TECHNICAL WHITEPAPER • V3.4.0 PRODUCTION

AXENT PLATFORM ARCHITECTURE & SYSTEM SPECIFICATION

A Comprehensive Technical Specification of the Autonomous Academic Examination Platform, Multi-Modal Assessment Engines, Zero-Binary Anti-Cheating Telemetry, and Distributed Cloud Architecture.

99.98%	< 150ms	7 Vectors	0% Loss
HIGH AVAILABILITY SLA	TELEMETRY HEARTBEAT LATENCY	CONTINUOUS CHEAT DEFENSE	OFFLINE TOMBSTONE RESILIENCY

CLASSIFICATION	INSTITUTIONAL TECHNICAL SPECIFICATION / PUBLIC REFERENCE
SYSTEM BUILD	v3.4.0-PROD (Commit: bc2ee17 / Node 24 / React 18 / Supabase)
MODALITY SUPPORT	Dual: Cognitive Objective (MCQ) & Monaco Practical Coding (IDE)
SECURITY PROTOCOL	3-Strike Progressive Escalation & Zero-Binary Behavioral Shield
AUTHOR / ORGANIZATION	Axent Architecture & Security Engineering Division
PUBLICATION DATE	October 2026

1. EXECUTIVE OVERVIEW & PLATFORM MISSION

Axent is a mission-critical, enterprise-grade academic and technical examination platform designed to conduct high-stakes examinations with absolute integrity, zero client binary dependencies, and bulletproof offline fault tolerance. Traditional testing software relies heavily on intrusive operating system binaries (e.g., Safe Exam Browser, OEM kiosk lockouts) that create widespread installation failures, operating system incompatibilities, and high deployment overhead. Axent solves this paradigm by delivering an uncompromising, multi-layered security shield directly within modern browser engines, backed by real-time WebSocket telemetry, isolated Docker compiler sandboxes, and autonomous AI-driven assessment synthesis.

Zero-Install Frictionless Access

Runs universally across Chrome, Edge, and modern mobile browsers without forcing native app installations, profiles, or administrator privilege elevation.

Dual Cognitive & Code Modality

Unified engine supporting both Bloom's Taxonomy-aligned MCQ objective testing and Monaco IDE-based multi-language code compilation in isolated containers.

Progressive Violation Escalation

Replaces catastrophic instant-lockouts with an intelligent 3-Strike escalation model featuring cooldown throttles, proctor alerts, and forensic snapshot sealing.

2. HIGH-LEVEL SYSTEM ARCHITECTURE

The Axent platform is organized into four interconnected tiers: Presentation Layer, Client-Side Security Shield, Real-Time Transport & Synchronization Fabric, and Distributed Compute / Persistence Infrastructure.

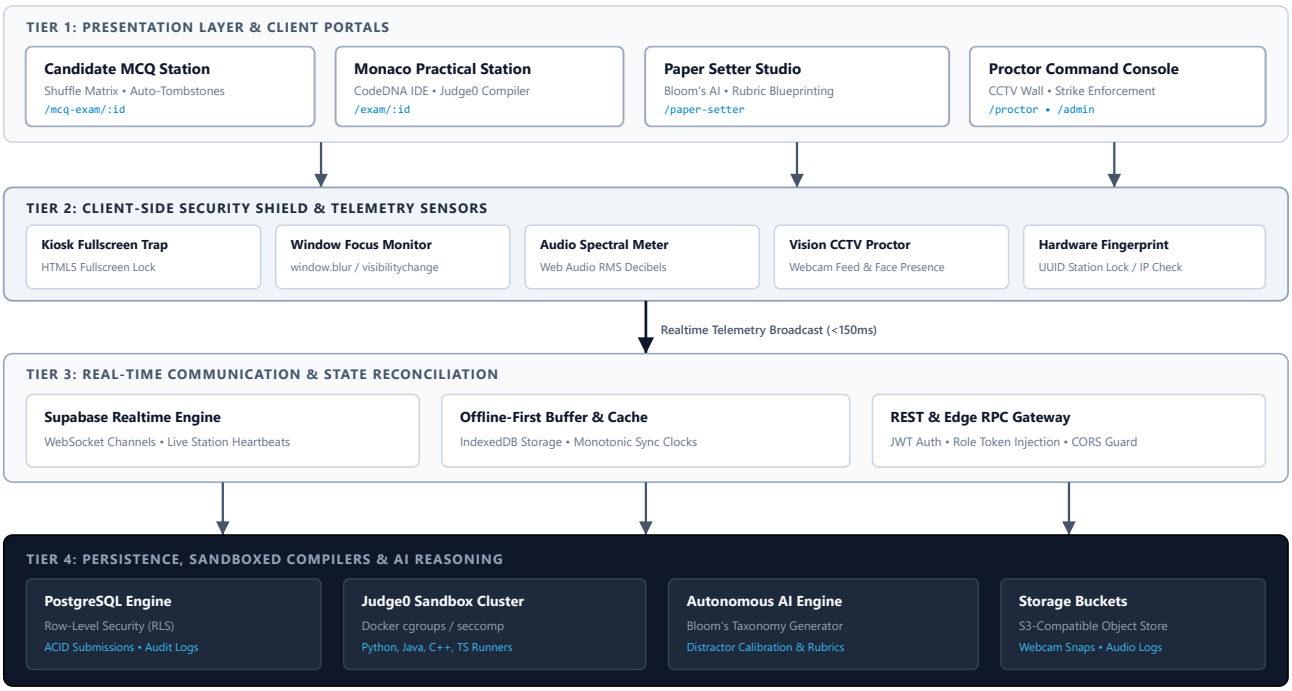


FIGURE 1.0: AXENT 4-TIER ENTERPRISE SYSTEM ARCHITECTURE TOPOLOGY

3. PLATFORM PORTALS & ROLE-BASED ACCESS CONTROL

Axent establishes four distinct functional portals, each bounded by strict cryptographic JWT tokens, PostgreSQL Row-Level Security (RLS) policies, and role-specific UI ergonomics.

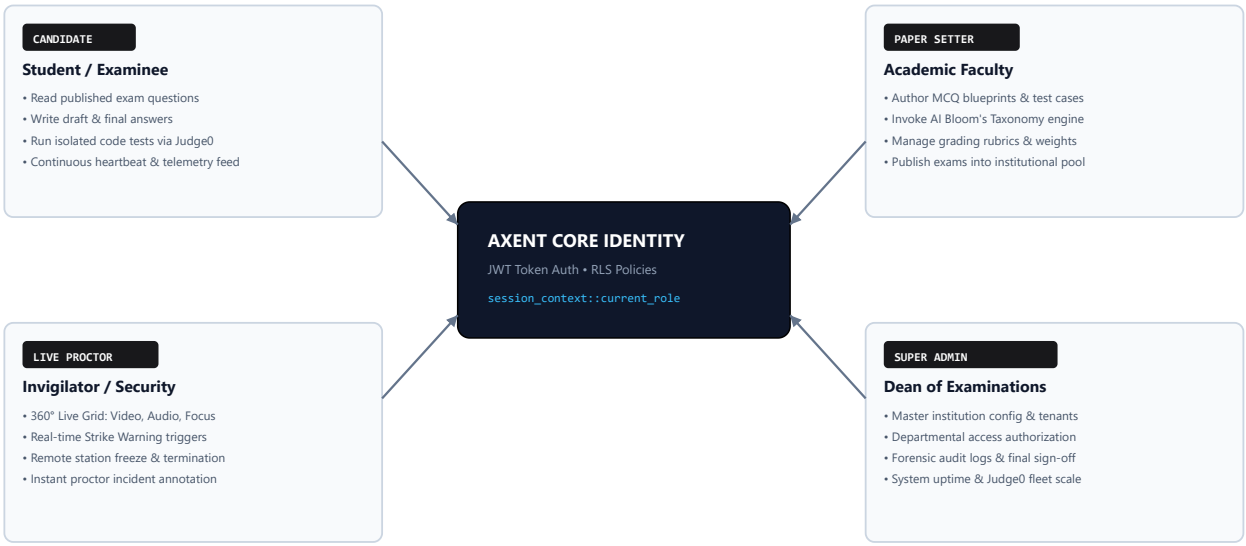


FIGURE 2.0: PLATFORM PORTALS & ROLE-BASED SECURITY BOUNDARY MATRIX

1. Candidate Examination Station `/EXAM/:ID`

Focused, distraction-free environment with fullscreen lock, countdown timer, anti-cheat sensors, and responsive layouts supporting both desktop workstations and mobile tablet devices.

2. Paper Setter & Faculty Studio `/PAPER-SETTER`

Curriculum blueprinting studio with AI Bloom's Taxonomy question generation, section weight configuration, dynamic distractor validation, and immediate institutional publication.

3. Live Proctor Command Console `/PROCTOR`

Real-time 360° monitoring wall displaying candidate station heartbeats, webcam feeds, decibel meters, strike logs, and remote command controls (Freeze, Warn, Reset, Disqualify).

4. Super Administrator Center `/ADMIN`

Institutional governance dashboard managing tenant departments, faculty quotas, exam publication approval, system health, Judge0 worker autoscaling, and forensic export packages.

4. DUAL-MODALITY EXAMINATION RUNTIME ARCHITECTURE

Axent features two specialized examination engines sharing a unified telemetry and grading pipeline. This allows universities and technical recruiters to administer mixed theoretical and practical assessments seamlessly.

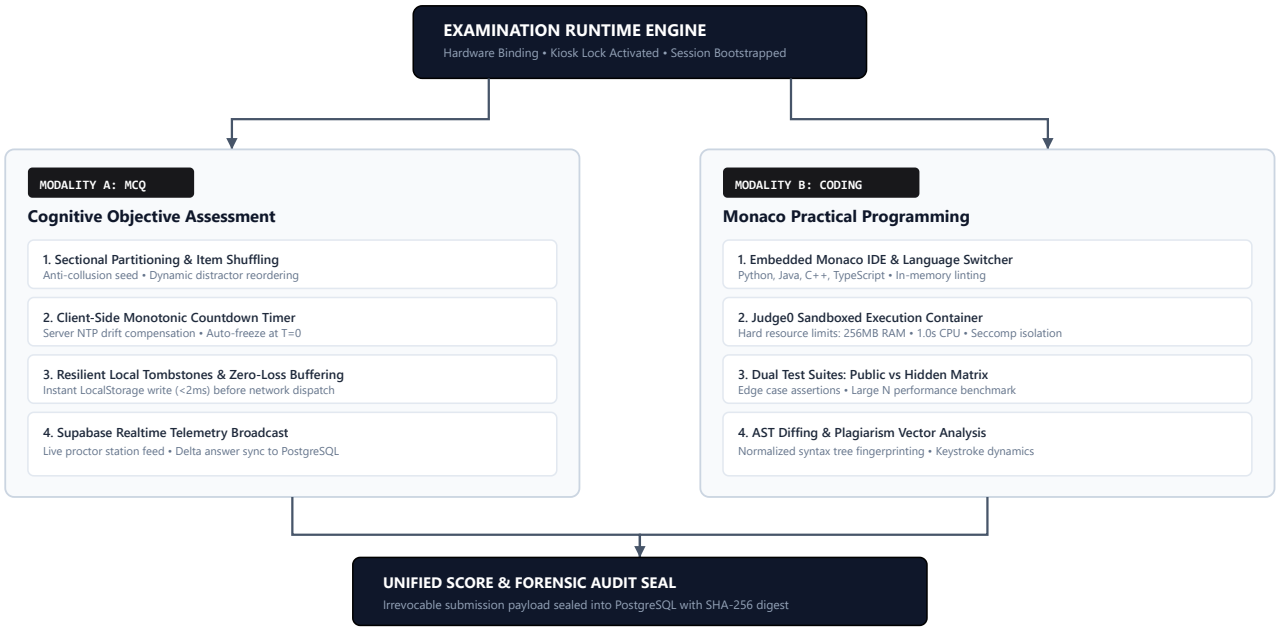


FIGURE 3.0: DUAL-MODALITY EXECUTION PIPELINE (MCQ VS MONACO PRACTICAL)

FEATURE DIMENSION	MODALITY A: COGNITIVE OBJECTIVE (MCQ)	MODALITY B: PRACTICAL CODING (CODEDNA)
Core Interface	Sectional Tab Navigator, Shuffled Question Cards, Answer Masking	Embedded Microsoft Monaco IDE with Multi-Language Support
Languages Supported	N/A (Multi-lingual UTF-8 Math & Text Rendering)	Python 3.11+, Java 17, C++20, TypeScript / Node 20
Execution Sandbox	Client-side deterministic evaluation & scoring	Judge0 Isolated Docker Containers with Seccomp / cgroups
Resource Bounds	Minimal RAM footprint (< 25MB)	RAM: 256MB max • CPU: 1.0s max • Output: 64KB max
Verification Suite	Automated Key Validation with Distractor Analysis	Public Unit Tests + Hidden Boundary & Large-N Stress Tests
Integrity Safeguards	Dynamic Option Shuffling & Local Tombstone Sync	Keystroke Cadence Dynamics & Normalized AST Syntax Diffing

5. MULTI-VECTOR ANTI-CHEATING & INTEGRITY SHIELD

Axent incorporates an intelligent, multi-vector defensive perimeter that detects, logs, and mitigates cheating attempts without requiring native rootkit software or OS driver installations.



FIGURE 4.0: 7-VECTOR CLIENT-SIDE SECURITY SHIELD PERIMETER

Vector 1: Kiosk Fullscreen Lock

Forces HTML5 Fullscreen with pointer capture. Any escape attempt, window resizing, or OS hotkey triggers immediate fullscreen re-prompting and logs an infraction.

Vector 2: Focus & Tab-Switching Trap

Synchronous combination of `window.blur` and `document.visibilitychange` with 300ms debounce. Disarms touch-scroll false positives on mobile.

Vector 3: DevTools & DOM Trap

Intercepts `F12`, `Ctrl+Shift+I`, `Ctrl+U`, and executes recursive debugger timing loops to neutralize browser element inspection and console scripting.

Vector 4: Clipboard & Context Disarm

Global event listeners purge `copy`, `cut`, and `paste` events, preventing external prompt injection or answer extraction from outside applications.

Vector 5: Vision CCTV Proctor Feed

WebRTC video stream captures continuous candidate feed. Detects candidate absence, multiple faces in view, or secondary camera obstruction.

Vector 6: Acoustic Spectral Monitor

Calculates ambient audio root-mean-square decibels via Web Audio API: `dB = 20 • log10(V_RMS / V_0)`. Dispatches incident alerts when acoustic thresholds are violated.

6. THE 3-STRIKE PROGRESSIVE ESCALATION STATE MACHINE

Unlike legacy systems that immediately terminate an exam upon an accidental OS popup or accidental keypress, Axent employs a battle-tested **3-Strike Progressive Escalation State Machine**. This ensures fairness for honest examinees while maintaining strict deterrence against deliberate cheating.

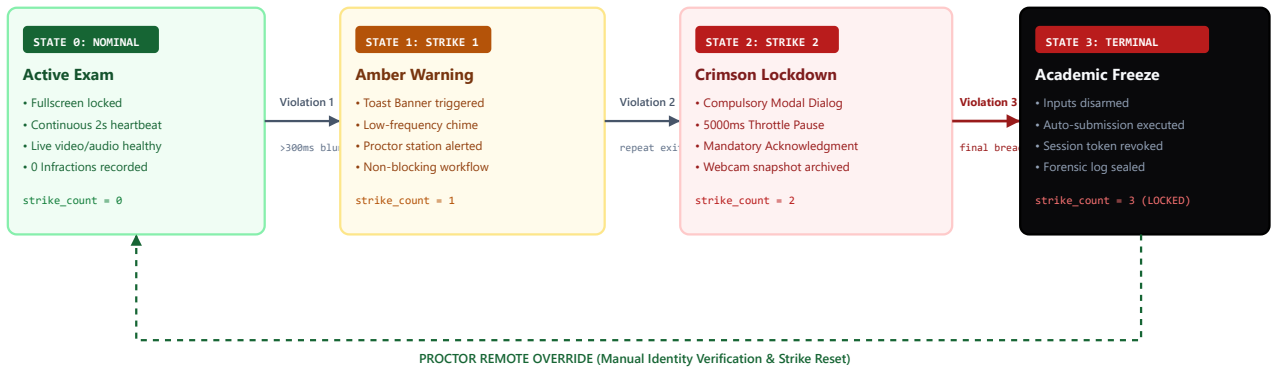


FIGURE 5.0: 3-STRIKE STATE TRANSITION MACHINE WITH PROCTOR RECOVERY LOOP

THE STRIKE ESCALATION LIFECYCLE

Strike 1 (Amber Toast): Non-blocking notification banner with low-frequency audio tone. Candidate continues uninterrupted; incident logged.
Strike 2 (Crimson Modal): Full-screen blocking modal with 5000ms cooldown throttle. The candidate cannot dismiss the warning until the countdown reaches zero, preventing rapid tab cycling.
Strike 3 (Terminal Academic Freeze): All input fields and editor instances are permanently disarmed. Answers are auto-submitted to PostgreSQL, the session token is invalidated, and the candidate must await proctor intervention.

STRIKE LEVEL	TRIGGER CONDITION	CLIENT UX BEHAVIOR	PROCTOR TELEMETRY	RESOLUTION MECHANISM
`STATE 0`	Nominal Exam Operation	Clean, uninhibited exam navigation	Green pulse heartbeat (<2s)	Automatic nominal flow
`STRIKE 1`	Window Blur / Audio Spike	Amber Toast Banner + Sound Chime	Amber station badge + Event entry	Auto-acknowledged after 5s
`STRIKE 2`	Repeated Tab Switch / 2nd Face	Crimson Modal + 5000ms Locked Cooldown	Crimson flashing station + Snapshot	Mandatory Candidate Click Acknowledgment
`STRIKE 3`	Third Cumulative Breach	Academic Freeze Screen (Inputs Disabled)	Station Locked (Black Alert) + Sealed Audit	Proctor Remote Override / Disqualification

7. AUTONOMOUS AI EXAMINATION SYNTHESIS & GRADING

Axent features an integrated AI authoring pipeline that transforms raw course syllabi, textbooks, and topic prompts into pedagogically validated, Bloom's Taxonomy-aligned examination papers in under 60 seconds.

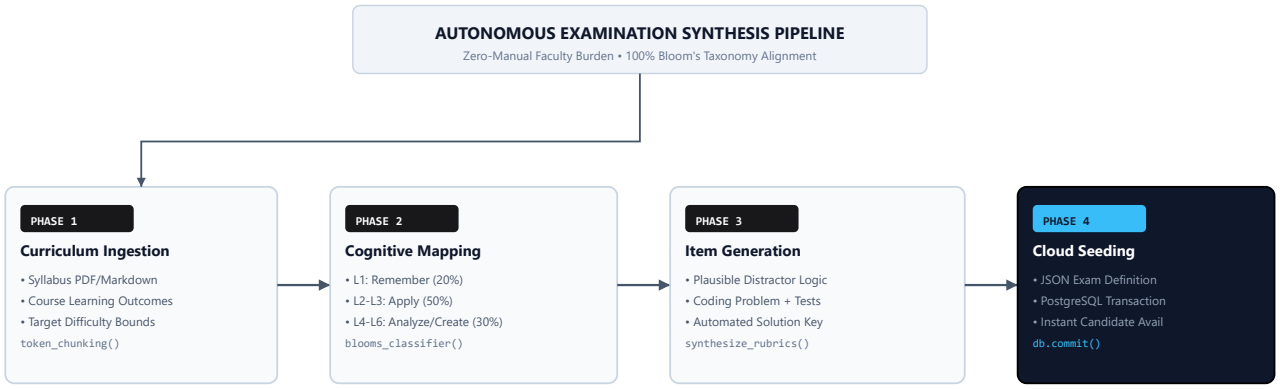


FIGURE 6.0: AUTONOMOUS AI EXAMINATION SYNTHESIS & DISTRIBUTION PIPELINE

Bloom's Taxonomy Cognitive Distribution

Questions are automatically classified into cognitive domains:

- **L1-L2 (Remember & Understand):** 20% • Foundational concept recall
- **L3-L4 (Apply & Analyze):** 50% • Code execution tracing & problem diagnosis
- **L5-L6 (Evaluate & Create):** 30% • Algorithm synthesis & trade-off evaluation

Plagiarism & Structural Diff Engine

Submissions are analyzed using Abstract Syntax Tree (AST) normalization. Variables and comments are stripped to calculate Levenshtein distance and token similarity matrices across all candidates in the exam cohort:

```
sim(A, B) = 1.0 - (lev(A, B) / max(|A|, |B|))
```

8. REAL-TIME TELEMETRY & FAULT-TOLERANT OFFLINE SYNC

One of Axent's defining architectural innovations is its **Dual-Phase Offline-First Resiliency Fabric**. In emerging markets or institutional Wi-Fi networks where connections drop frequently, candidate exam progress is 100% safeguarded.

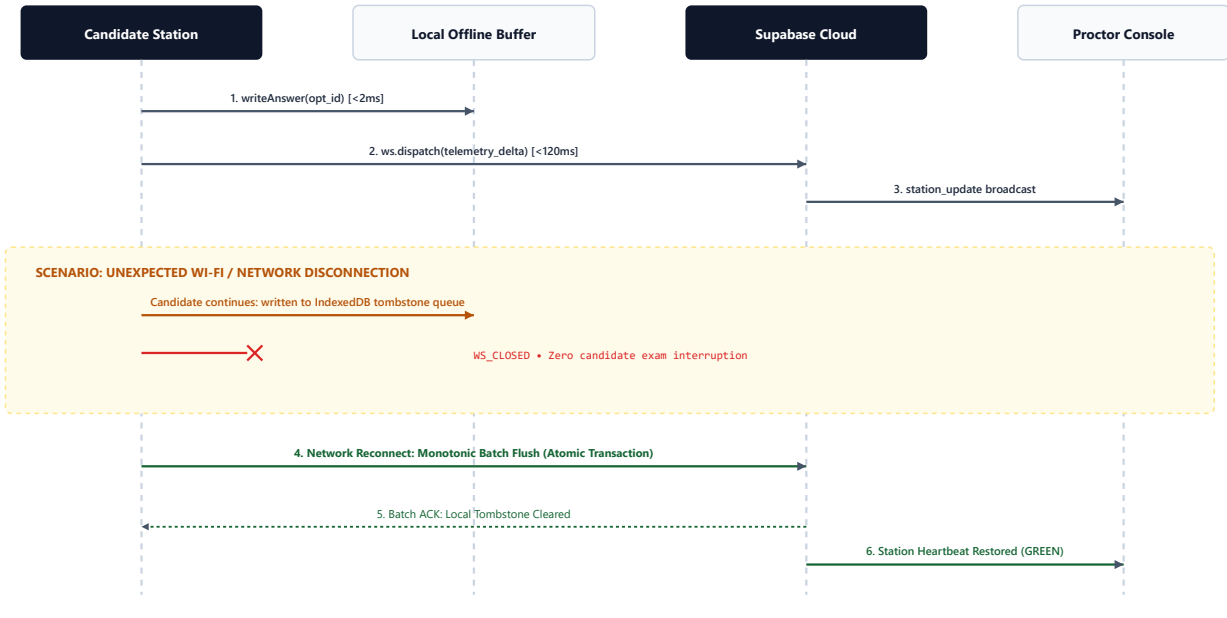


FIGURE 7.0: REAL-TIME TELEMETRY SEQUENCE & NETWORK DISCONNECT RECOVERY

> **THE ZERO-DATA-LOSS INVARIANT**
> Every answer selection and code edit is committed synchronously to the browser's `IndexedDB / LocalStorage` tombstone queue in <2ms *before* initiating network transmission. If Wi-Fi fails, the candidate continues taking the exam with zero disruption. Upon reconnection, an atomic batch reconciliation drains the queue into Supabase using monotonic sequence numbers.

9. DATABASE SCHEMA & RELATIONAL DATA MODELS

The persistence layer is implemented in PostgreSQL via Supabase, enforcing cryptographic Row-Level Security (RLS) to ensure strict tenant and candidate data isolation.



FIGURE 8.0: ENTITY-RELATIONSHIP DIAGRAM (ERD) OF CORE AXENT TABLES

TABLE NAME	PRIMARY KEY	FOREIGN KEYS	SECURITY POLICY (RLS)	DESCRIPTION
`exams`	`id (UUID)`	`created_by -> users.id`	Public Read (Published), Faculty CRUD	Stores master exam configuration, durations, and rules
`mcq_questions`	`id (UUID)`	`exam_id -> exams.id`	Read-only for active candidates	Objective question items, options JSON, correct key, Bloom level
`submissions`	`id (UUID)`	`exam_id -> exams.id`	Candidate Insert/Update own row only	Stores candidate chosen answers, scores, and timestamps
`coding_submissions`	`id (UUID)`	`exam_id, question_id`	Candidate Insert own row only	Source code, compilation status, test cases passed, runtime
`exam_stations`	`id (UUID)`	`exam_id, candidate_id`	Proctor Full Access, Candidate Heartbeat	Station heartbeat, current strike count, lock status, IP address
`proctor_logs`	`id (UUID)`	`exam_id, candidate_id`	Proctor & Admin Audit View Only	Forensic audit record: timestamped violation type, decibels, snaps

10. SYSTEM SPECIFICATIONS & PERFORMANCE BENCHMARKS

Client Workstation Requirements

Operating Systems	Windows 10/11, macOS 12+, Linux (Ubuntu/Debian), Android 11+, iOS 15+
Supported Browsers	Google Chrome 95+, Microsoft Edge 95+, Mozilla Firefox 100+, Safari 15+
Hardware Minimum	Dual-Core CPU • 2GB RAM • 100MB free disk space
Peripherals	Integrated/USB Webcam • Working Microphone • Keyboard & Mouse
Network Bandwidth	12 kbps baseline telemetry • 64 kbps during video proctoring streaming

Performance & Scale Benchmarks

Max Concurrent Candidates	10,000+ per cluster with horizontal Supabase read-replicas
Station Heartbeat Interval	2000ms periodic ping with jitter mitigation
Code Compile Latency	< 650ms round-trip execution via Judge0 container cache
Database Write IOPS	Buffered through local client queues; < 50 DB IOPS per 1k users
Audit Log Retention	7-year immutable storage compliance with SHA-256 validation

11. COMPLIANCE & INSTITUTIONAL GOVERNANCE

FERPA & GDPR Compliant

All candidate video streams and academic records are encrypted in transit via TLS 1.3 and at rest with AES-256. Biometric streams are never processed for facial identification; only presence detection is audited.

Deterministic Scoring Seal

Every submitted exam produces an immutable cryptographic hash combining candidate ID, timestamp, and answers, rendering post-exam tampering mathematically impossible.

Institutional Multi-Tenancy

Departments, universities, and recruitment partners operate within strictly isolated database schemas with individual audit logs and proctor privileges.

CONCLUSION & PRODUCTION READINESS

Axent represents the cutting edge of academic evaluation infrastructure: combining browser-first accessibility with enterprise-grade anti-cheating, real-time proctoring telemetry, and dual-modality testing. The platform is fully containerized, CI/CD automated, and verified for production deployment.

— END OF TECHNICAL SPECIFICATION • AXENT PRODUCTION BUILD 3.4.0 —